

DM PROGECT

1. Cluster Profiling

Cluster 0:

- Recency ≈ 41 (recent customers)
- Frequency ≈ 4.8 (moderate purchases)
- Monetary ≈ 2012 (medium spending)

Cluster 1:

- Recency ≈ 246 (inactive customers)
- Frequency ≈ 1.5 (very low purchases)
- Monetary ≈ 631 (low spending)

Cluster 2:

- Recency ≈ 7 (very recent)
- Frequency ≈ 80 (extremely frequent)
- Monetary $\approx 122,888$ (very high spending)

2. Customer Personas

Cluster 0 → "Regular Customers"

- Buy occasionally
- متوسطين في كل حاجة -

Cluster 1 → "Lost / Low-Value Customers"

- مش بيشتروا بقالهم فترة كبيرة -
- Spending جدًا قليل -

Cluster 2 → "VIP Customers"

- بيشتروا كثير جدًا -
- بيصرفوا فلوس ضخمة -

3. Business Recommendations

Cluster 0:

- Offer promotions to increase frequency
- Upsell products

Cluster 1:

- Send reactivation campaigns
- Offer discounts to bring them back

Cluster 2:

- Provide loyalty programs
- Give exclusive offers
 - Focus on retention (very important group)

4. Cluster Size Distribution

- Cluster 0: 3231 customers (~74%)
- Cluster 1: 1093 customers (~25%)
- Cluster 2: 14 customers (~0.3%)

Insight:

- معظم العملاء عاديين (Cluster 0)
- قليل جدًا لكن مهم جدًا VIP

5. Cluster Interpretation

- Cluster 0: Active average customers
- Cluster 1: Inactive and low-value customers
- Cluster 2: Extremely valuable customers (VIP)

6. Cluster Naming

- Cluster 0 → "Regular Customers"
- Cluster 1 → "Churned Customers"
- Cluster 2 → "VIP Customers"

7. Business Insights

- A very small group of customers (Cluster 2) generates the highest revenue
- A large portion of customers are inactive or low-value (Cluster 1)
- Most customers fall into the average segment (Cluster 0)

The business relies heavily on a small number of high-value customers

8.Final Explanation

This project applied RFM analysis and KMeans clustering to segment customers based on their purchasing behavior.

The results show that:

- Customers can be divided into 3 clear groups
- A very small group (VIP) generates extremely high revenue
- Many customers are inactive or low-value

Business Value:

- Helps target customers effectively
- Improves marketing strategies
- Increases customer retention
- Identifies high-value customers

Overall, this model helps transform raw data into actionable business decisions.